

PAPER_1557 — Higgs Boson Mass = $m_H = 2 \cdot A_5 + N_{CH} - D_{phys} + F_{TRZ} \cdot SSQ + F_{TRZ}^2 \cdot D_{BSFG} + F_{TRZ}^2 \cdot SSQ^2 \approx 125.12 \text{ GeV}$

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Particle Physics **Date:** June 18, 2026 **Status:** CLOSED — 0.016% closure from PAPER_1209HH_S656

Observation

PAPER_1209HH_S656 (Particle Physics Unified Proof Set) lists **Higgs Boson Mass** with the integer-primitive expression:

Higgs Boson Mass = $m_H = 2 \cdot A_5 + N_{CH} - D_{phys} + F_{TRZ} \cdot SSQ + F_{TRZ}^2 \cdot D_{BSFG} + F_{TRZ}^2 \cdot SSQ^2 \approx 125.12 \text{ GeV}$
Status: **0.016%**.

UQFF Closed Identity

$m_H = 2 \cdot A_5 + N_{CH} - D_{phys} + F_{TRZ} \cdot SSQ + F_{TRZ}^2 \cdot D_{BSFG} + F_{TRZ}^2 \cdot SSQ^2 \approx 125.12 \text{ GeV} \quad 0.016\%$

Physical Interpretation

Clean integer-primitive form at first order: $2 \cdot A_5 + N_{CH} - D_{phys} = 120 + 9 - 4 = 125 \text{ GeV}$ — exactly the Higgs boson mass to 3 significant figures, before small F_{TRZ} corrections refine to 125.12 vs CODATA 125.10.

NOT REPLACEMENT

CODATA/PDG treats Higgs Boson Mass as a precision-measured constant. UQFF supplies the structural integer-primitive form, demonstrating the framework's predictive economy without claiming to replace experimental measurement.

Reference

- Source: PAPER_1209HH_S656
- Related: PAPER_1209 series unified proof sets
- Calculator dispatch: `calculate_paradox({"paradox": "m_h_higgs_125_10"})`

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